

## Reviewed Preprint

Published from the original preprint after peer review and assessment by eLife.

[About eLife's process](#)

## Reviewed preprint version 1

March 19, 2024 (this version)

## Posted to preprint server

October 9, 2023

## Sent for peer review

September 28, 2023

# Machine learning and biological validation identify sphingolipids as key mediators of paclitaxel-induced neuropathy in cancer patients

Jörn Lötsch, Khayal Gasimli, Sebastian Malkusch, Lisa Hahnefeld, Carlo Angioni, Yannick Schreiber, Sandra Trautmann, Saskia Wedel, Dominique Thomas, Nerea Ferreiros Bouzas, Christian Brandts, Benjamin Schnappauf, Christine Solbach, Gerd Geisslinger, Marco Sisignano 

Institute of Clinical Pharmacology, Goethe - University, Theodor Stern Kai 7, 60590 Frankfurt am Main, Germany • Fraunhofer Institute for Translational Medicine and Pharmacology ITMP, and Fraunhofer Cluster of Excellence for Immune Mediated Diseases CIMD, Theodor-Stern-Kai 7, 60596 Frankfurt am Main, Germany • Goethe University, Department of Gynecology and Obstetrics, Theodor-Stern-Kai 7, Frankfurt/Main, Germany • German Cancer Consortium (DKTK) and German Cancer Research Center (DKFZ), Heidelberg, 69120, Germany • Goethe University, University Cancer Center Frankfurt (UCT), Goethe University Hospital, Frankfurt am Main 60590, Germany • Oncology Center, Sana-Klinikum Offenbach, Starkenburgring 66, 63069 Offenbach

 [https://en.wikipedia.org/wiki/Open\\_access](https://en.wikipedia.org/wiki/Open_access)

 Copyright information

## Abstract

## Background

Chemotherapy-induced peripheral neuropathy (CIPN) is a serious therapy-limiting side effect of commonly used anticancer drugs. Previous studies suggest that lipids may play a role in CIPN. Therefore, the present study aimed to identify the particular types of lipids that are regulated as a consequence of paclitaxel administration and may be associated with the occurrence of post-therapeutic neuropathy.

## Methods

High resolution mass spectrometry lipidomics was applied to quantify  $d = 255$  different lipid mediators in the blood of  $n = 31$  patients drawn before and after paclitaxel therapy for breast cancer treatment. A variety of supervised statistical and machine-learning methods was applied to identify lipids that were regulated during paclitaxel therapy or differed among patients with and without post-therapeutic neuropathy.

## Results

Twenty-seven lipids were identified that carried relevant information to train machine learning algorithms to identify, in new cases, whether a blood sample was drawn before or

after paclitaxel therapy with a median balanced accuracy of up to 90%. One of the top hits, sphinganine-1-phosphate (SA1P), was found to induce calcium transients in sensory neurons via the transient receptor potential vanilloid 1 (TRPV1) channel and sphingosine-1-phosphate receptors. SA1P also showed different blood concentrations between patients with and without neuropathy.

## Conclusions

Present findings suggest an important role for sphinganine-1-phosphate in paclitaxel-induced biological changes associated with neuropathic side effects. The identified SA1P, through its receptors, provides a potential drug target for co-therapy with paclitaxel to reduce one of its major and therapy-limiting side effects.

### eLife assessment

Sisigano et al. report findings about the role of sphingolipids using lipidomics with machine learning in paclitaxel-induced peripheral neuropathy and preliminary translation of the impact of SA1P in cultured neuronal cells. This study presents a **valuable** finding on the increased activity of two well-studied signal transduction pathways in a subtype of breast cancer. The strength of evidence is **incomplete** with some support for the main claims with some limitations.

## Introduction

Paclitaxel is a standard adjuvant treatment for breast cancer and several other cancers. Originally isolated from the yew tree *Taxus brevifolia*, it inhibits mitosis by stabilizing microtubules and preventing tubulin depolymerization (5 [1](#)–7 [2](#)). A serious dose- and therapy-limiting side effect, which it shares with other commonly used cytostatic drugs, is the chemotherapy-induced peripheral neuropathy and neuropathic pain (CIPN), which affects up to 80% of treated patients (1 [3](#), 2 [4](#)). Currently, there are no pharmacologic treatments for CIPN except for duloxetine (1 [5](#), 3 [6](#), 4 [7](#)). Therefore, research on the mechanism of paclitaxel induced CIPN with possible identification of novel treatments is an active research topic.

Several genes and neurofilament proteins have been implicated in paclitaxel-induced neuropathy. More recently, lipid mediators have been shown to be produced at high levels in sensory neurons, neuronal tissue and immune cells after chemotherapy due to oxidative stress and have been shown to contribute to chemotherapy-induced neuropathy and neuropathic pain by modulating neuronal ion channels (11 [8](#)–14 [9](#)). Therefore, they are of particular interest as signaling molecules and potential markers for chemotherapy-induced neuropathy in patients. In fact, lipids are already considered markers for other neurological diseases such as Alzheimer's disease and amyotrophic lateral sclerosis (ALS) (15 [10](#)–17 [11](#)). However, a systematic approach to investigate lipids in patients with paclitaxel-induced neuropathy has not been performed.

In this prospective clinical cohort study, plasma concentrations of 255 different lipid mediators were evaluated for changes in their concentrations associated with paclitaxel treatment. A comprehensive LC-MS/MS-based targeted and LC-QTOFMS-based untargeted lipidomics screening was performed on plasma samples from paclitaxel-treated patients. Lipid groups measured included eicosanoids, endocannabinoids, oxidized linoleic acid metabolites, sphingolipids,

lysophospholipids and free fatty acids, many of which have been previously associated with persistent pain states (18–20). A data-driven approach was used to identify lipid mediators whose concentrations could be used to train machine learning algorithms to identify, in new cases, whether a plasma sample was collected before or after therapy, or from a patient with or without post-therapy neuropathy. The biological relevance of the findings was then validated in vitro by applying SA1P to primary sensory neurons using calcium imaging.

## Materials and Methods

### Patients and study design

This was a prospective single-arm study enrolling patients with breast cancer. The study was conducted in accordance with the Declaration of Helsinki on Biomedical Research Involving Human Subjects and was approved by the Ethics Committee of the Medical Faculty of the Goethe-University, Frankfurt am Main, Germany. Informed written consent was obtained from each of the participants.

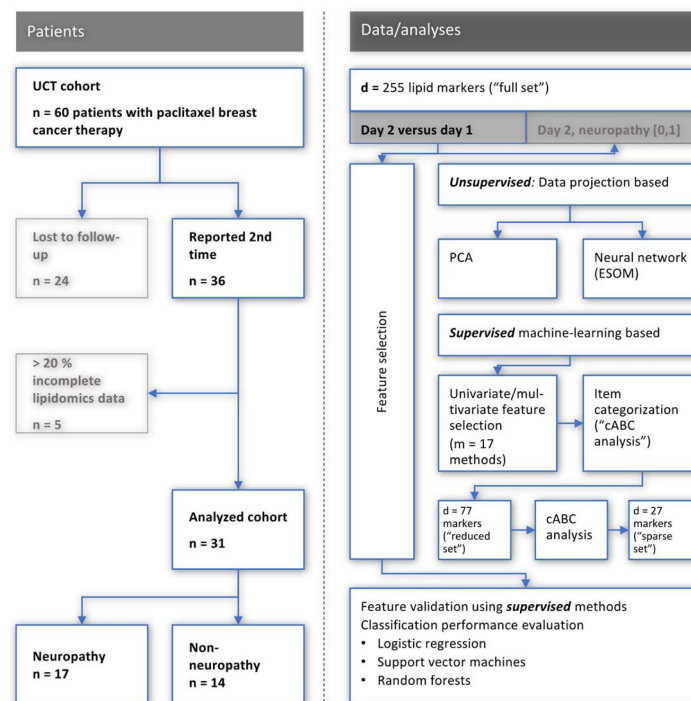
Sixty patients (one male, 59 female with breast cancer and undergoing paclitaxel treatment were recruited from the Tumor Center of the University Hospital Frankfurt, Germany (UCT). Most patients ( $n = 47$ ) received the “paclitaxel-weekly” schedule, consisting of 12 cycles of paclitaxel treatment ( $80 \text{ mg/m}^2$ , each week); a few patients ( $n = 13$ ) received mixed carboplatin/paclitaxel treatment (Table S1). A blood sample was collected from each patient before and after chemotherapy, and the degree of neuropathy after chemotherapy was assessed as described below. Plasma was isolated from the blood samples immediately after blood collection to ensure lipid stability. Plasma was stored at  $-80^\circ\text{C}$  until analysis.

All patients provided a blood sample before chemotherapy; however, only 36 patients of the patients provided a second blood sample after chemotherapy. For our analysis, we focused on patients that had two blood samples. Therefore, 72 samples from 36 patients (before and after chemotherapy) were analyzed using both LC-MS/MS-based targeted and LC-QTOFMS-based untargeted lipidomics. A total of 255 individual lipids were detected in each sample. From the resulting data, five patients had to be excluded due to incomplete lipidomics data (more than 20% of the analytes could not be detected). The remaining 62 samples from 31 patients were used for the machine learning analysis (Figure 1).

### Assessment of neuropathy

The occurrence and severity of peripheral neuropathy was assessed according to the guidelines of the NCI Common Terminology Criteria for Adverse Events (CTCAE) v5.0, Published: November 27, 2017, by the U.S. Department of Health and Human Services ([https://ctep.cancer.gov/protocoldevelopment/electronic\\_applications/docs/ctcae\\_v5\\_quick\\_reference\\_5x7.pdf](https://ctep.cancer.gov/protocoldevelopment/electronic_applications/docs/ctcae_v5_quick_reference_5x7.pdf)). Neuropathy assessment was performed prospectively, i.e., before the first paclitaxel treatment and again after the 12<sup>th</sup> treatment cycle. Neuropathy was assessed regularly upon visit of the patient. The last assessment was performed 4.5 years after initial chemotherapy (Table S1).

The severity of neuropathy was graded into five grades: Grade 1 Mild; asymptomatic or mild symptoms; clinical or diagnostic observations only; intervention not indicated. Grade 2 Moderate; minimal, local or noninvasive intervention indicated; limiting age-appropriate instrumental ADL (activities of daily living). Grade 3 Severe or medically significant but not immediately life-threatening hospitalization or prolongation of hospitalization indicated; disabling; limiting self-care ADL. Grade 4 Life-threatening consequences; urgent intervention indicated. Grade 5 Death



**Figure 1**

### Flowchart showing the number of patients included and the workflow of the data analysis

UCT: University Cancer Center Frankfurt, PCA: principal component analysis, ESO: emergent self-organizing maps, cABC analysis: computed ABC analysis. The figure was created using Microsoft PowerPoint® (Redmond, WA, USA) on Microsoft Windows 11 running in a virtual machine powered by VirtualBox 6.1.36 (Oracle Corporation, Austin, TX, USA) as guest on Linux, and then further modified with the free vector graphics editor "Inkscape (version 1.2 for Linux, <https://inkscape.org/>)

related to AE. In the present cohort, grades 1-3 were detected following paclitaxel chemotherapy (Table S1). Of the 31 patients with a full set of samples, 17 had neuropathy after chemotherapy (54.9%), 12 had grade 1, 3 had grade 2 and two patients with grade 3 (Table S1).

## Lipidomics analysis using LC-MS/MS and LC-QTOFMS

Blood was collected from patients in EDTA tubes and immediately centrifuged at 2000 xg for 10 minutes at 4°C. The supernatant was immediately frozen at -80°C until further processing. Approximately 2-3 ml of plasma was considered sufficient for each patient. Liquid chromatography-tandem mass spectrometry (LC-MS/MS) analysis of eicosanoids, oxidized linoleic acid metabolites (O(x)LAMs), prostanoids, endocannabinoids, LPAs, pterins, sphingolipids and ceramides, and lipidomics screening were performed as described previously (13, 19, 21). A total of 255 lipids were quantified in each plasma sample. These lipids belong to the groups of eicosanoids, oxidized linoleic acid metabolites, endocannabinoids, lysophosphatidic acids, pterins, sphingolipids, ceramides, cholesterol, cholesterol esters, diacylglycerols, triacylglycerols, phospholipids, lysophospholipids, and free fatty acids. Full details of the lipids detected are given in Table S2. Full details of the LC-MS methods used can be found in the Supporting Information Methods section (Tables 1-10).

## Data analysis

The data analysis combined statistical and machine learning methods (“mixture of experts” approach) shown to be superior to relying on a single method (23)(24), such as regression analysis alone (22). The data analysis included unsupervised and supervised methods (Figure 1). Unsupervised methods were used to establish whether the lipidomics data contained structures that supported prior classification into (i) baseline and post-treatment samples, or (ii) subjects with or without neuropathy in the post-treatment samples. Supervised methods were then used to identify lipid mediators that carried information relevant to the class structure of the data set. Programming was performed in R language (25) using the R software package (26), version 4.1.2, for Linux, available free of charge from the Comprehensive R Archive Network (CRAN) at <https://CRAN.R-project.org/>, and in Python language (27) using Python version 3.8.12, available free of charge at <https://www.python.org> (accessed March 1, 2022). The data analysis is summarized in Figure 1 and described in detail in the supporting information.

Following log-transformation and missing-value imputation, the lipid marker concentrations were analyzed using unsupervised methods to assess whether they contained structures that were consistent with the prior class structure. Z-standardized data were projected from the high-dimensional space onto lower dimensional planes by means of principal component analysis (PCA) (28, 29) as one of the classical methods established. As an alternative approach, unsupervised machine learning was implemented as self-organizing maps (SOM) of artificial neurons (30). In their special form of emergent SOM (ESOM (31)), the present map consisted of 4,000 neurons arranged on a two-dimensional toroidal grid with 50 rows and 80 columns (32, 33).

Supervised methods included binary logistic regression (36) and, since a mixture of experts was preferred (22-24), additional machine-learning algorithms of different types known to work well for tabular numerical data were applied. These included random forests (37, 38) as a robust tree-based bagging classifier, support vector machines (SVM) (39) as a hyperplane separation-based method. Classifier tuning is described in the supporting information. The analyses were performed using cross-validation and training, testing, and validation splits of the dataset as a machine learning standard. That is, a class-proportional random sample of 20% of the dataset was set aside before the analyses started and served as a validation sample that was not touched during algorithm training and feature selection. In the remaining 80% of the dataset, feature selection was performed using supervised analyses in 100-fold nested cross-validation scenarios. Feature selection, i.e., identification of the individual lipid mediators that contained

relevant information to assign a sample to either the pre- or post-therapy day or to a patient with or without neuropathy, was identified by majority vote among 17 different feature selection methods, including univariate and multivariate types listed in the supporting information. After quantifying the importance of each feature, the most relevant subset of lipid mediators was identified by subjecting the importance measures to computed ABC analysis (cABC analysis) (40 [↗](#)), an item categorization technique adopted from economics that aims to divide a set of positive numerical data into three disjoint subsets labeled “A,” “B,” and “C”. The set “A” should contain the “few important” elements (41 [↗](#)), which was retained. The resulting final sets of lipid mediators were then used to train the algorithms to perform the task of assigning a sample to either the pre- or post-therapy day or to a patient with or without neuropathy in the hold-out 20% validation sample mentioned above.

## Calcium Imaging with primary sensory neurons

Primary sensory neurons were cultured as described previously (42 [↗](#)). For calcium imaging experiments, neurons were stained with Fura-2-AM (Thermo Fisher) for at least 60 min at 37°C and washed afterwards twice with Ringer’s solution consisting of 145 mM NaCl, 1.25 mM  $\text{CaCl}_2 \times 2\text{H}_2\text{O}$ , 1 mM  $\text{MgCl}_2 \times 6\text{H}_2\text{O}$ , 5 mM KCl, 10 mM D-glucose, and 10 mM HEPES adjusted to a pH of 7.3. To investigate the effect of SA1P or LPC 24:0 on different TRP channels, sensory neurons were incubated with the lipids for 1 min at a concentration of 1 or 10  $\mu\text{M}$ , respectively. The gold standard agonists for TRPV1 and TRPA1 were capsaicin (200 nM, 20s) and AITC (allyl isothiocyanate, 75  $\mu\text{M}$ , 30s). Fingolimod was used at a concentration of 1  $\mu\text{M}$  and pre-incubated for 1 h prior to measurement. As a positive control, final stimulation with KCl (50 mM, 1 min) was used to depolarize all neurons. All stimulating compounds were dissolved in Ringer’s solution to their final concentrations.

The calcium imaging data were analyzed using descriptive statistics. All calcium imaging data are presented as the mean  $\pm$  SEM. Normal distribution was confirmed using the Shapiro-Wilk test. For experiments comparing only two groups, unpaired and heteroscedastic Student’s t-tests were conducted following Welch’s correction. When comparing more than two groups, one-way analysis of variance (ANOVA) was used, and for the comparison of more than three groups, two-way ANOVA was conducted. For all statistical analyses of the calcium imaging data, the software GraphPad Prism 9.5 was used. Statistical significance was set at p value < 0.05.

## Results

All patients received paclitaxel as adjuvant or neoadjuvant therapy, without any other potentially neurotoxic substances. Of the 60 patients from our analysis cohort, two were excluded due to rescheduling of paclitaxel therapy. Blood samples were obtained from 31 patients before and after chemotherapy. Twenty lipid marker variables had > 20% missing values. Following exclusion of these patients and variables and imputation, a data matrix for further analyses was obtained, sized  $79 \times 255$  (79 data set instances, samples, and 255 different lipid mediators **Figure 1** [↗](#)). An overview of the distribution of the lipid marker data is shown in Figure S1. These included 48 samples drawn on day 1 before therapy and 31 samples drawn on day 2 after 12 cycles of paclitaxel therapy. On day 2,  $n = 17$  of the 31 patients had symptoms of neuropathy (54.9%), which is in line with previous clinical reports on occurrence and severity of paclitaxel-induced neuropathy (43 [↗](#), 44 [↗](#)). Most patients reported grade 1 neuropathy, although two patients experienced grade 3 neuropathy. Occurrence and degree of neuropathy were monitored 4.5 years after finishing chemotherapy. The neuropathy lasted for several months or, in many cases, still persisted for 4.5 years after chemotherapy at the last examination (Table S1).

## Results of unsupervised analysis of structure in the lipidomics data supporting prior knowledge

An overview of the distribution of the lipid marker data is shown in Figure S1. PCA yielded  $d = 28$  components with eigenvalues  $> 1$ , which together explained 93.93% of the total variance in the lipid mediators (Figure 2). The  $d = 86$  lipid mediators that contributed most to the relevant PCs were identified based on the membership to category “A” in the cABC analysis of the weighted variable contributions to each PC (Figure S2). This was carried over as one of the several feature importance measures to the supervised analyses reported in the next chapter. On the emergent self-organizing map (ESOM, Figure 2a), a clear separation of two clusters was observed, which provided support that the lipid mediators contained a data structure contingent with the prior classification into pre- and posttherapy samples (Fisher’s exact test:  $p = 0.01054$ , odds ratio: 3.572973 with 95% confidence interval 1.28 - 10.52; Figure 2b). The separation of samples on the ESOM also corresponded, but to a lesser extent, with the occurrence of neuropathy observed at the time of the 2<sup>nd</sup> blood sample ( $p = 0.03279$ , odds ratio 0.16, 95% confidence interval 0.0137 - 1.022).

The results of the unsupervised analysis thus supported that the lipidomics data contained a structure contingent on a known prior classification. This supported the continuation of data analysis with supervised methods to determine which of the lipids carried relevant information to assign a probe to a particular prior class.

## Results of supervised analyses identifying lipid mediators relevant to the class structure

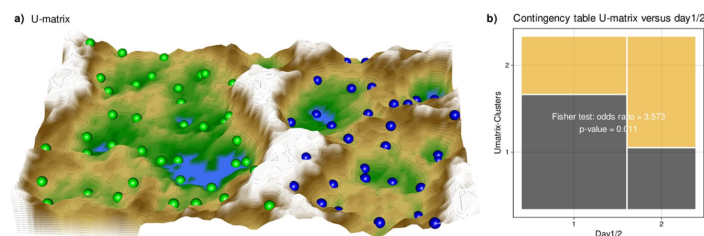
### Lipid mediators informative for assigning samples to before or after paclitaxel therapy

Based on the majority vote of the different approaches to feature selection including PCA importance and further univariate and multivariate feature selection methods specified in the supplementary information,  $d = 77$  lipid mediators were found to provide relevant information on whether a sample was collected before or after paclitaxel therapy (Figure S3). When statistical (logistic regression) and machine learning (random forests, support vector machines - SVM) algorithms were trained with this set of lipid mediators, the assignment of a sample to day 1 or 2 was well above the guessing level (Table 2). By contrast, when the training data were randomly permuted, the performance fell to a balanced accuracy of 0.5, i.e., guessing level, which established that the obtained class assignment in the non-permuted scenario had not been due to overfitting. In addition, by rerunning the cABC analysis on the mediators assigned to subset “A” in the first run (“recursive” cABC analysis (45)), the informative set of lipid mediators could be further reduced to  $d = 27$  (Table 1). With these mediators, SVM and random forest were still able to detect whether an instance of a lipidomics dataset was from before or after paclitaxel treatment at a balanced accuracy better than expected from guessing.

### Lipid mediators informative for assigning post-paclitaxel therapy samples to neuropathy

The  $n = 31$  samples from day 2 were probably too small to detect whether a sample was from a patient with neuropathy. Although the median balanced accuracies for the three algorithms to detect neuropathy in new cases unseen during training were up to 0.75, while only the guess level of 0.5 was achieved when using permuted data for training, the 95% CI of the performance measures was not separated from guess level. Therefore, features were selected using univariate methods (Cohen’s  $d$ , FPR, FWE). The small set of lipid mediators emerging from all three methods as informative for neuropathy again included SA1P (see above), and in addition sphingomyelin 33:1, and sphingomyelin 43:1. The three mediators differed significantly between samples from neuropathy-positive and neuropathy-negative patients (Figure 3).





**Figure 2**

## Results of a projection of the z-standardized log-transformed lipidomics data onto a lower-dimensional space by means of a self-organizing map of artificial neurons (bottom)

**a):** Emergent self-organizing map (ESOM), providing a 3-dimensional U-matrix visualization of distance-based structures of the serum concentrations of  $d = 255$  lipid mediators following projection of the data points onto a toroid grid of 4,000 neurons where opposite edges are connected. The dots represent the so-called “best matching units” (BMU), i.e., neurons on the grid that after ESOM learning carried a data vector that was most similar to a data vector of a sample in the data set. Please note that one BMU can carry vectors of several cases, i.e., the number of BMUs is not necessarily equal to the number of cases. The U-matrix visualization was colored as a top view of a topographic map with brown (up to snow-covered) heights and green valleys with blue lakes. Watersheds indicate borderlines between two different regions (clusters) separated by the white “mountain ridge” at the left of the U-matrix. BMUs belonging to the two different clusters are colored in green or bluish. **b):** Mosaic plot of the prior classes (day 1 or day 2) versus the ESOM/Umatrix based clusters. The separation corresponded to the prior classification into pre- and posttherapy probes (day1/2). The figure has been created using the R software package (version 4.1.2 for Linux; <https://CRAN.R-project.org/> (26)) and the libraries “ggplot2” (<https://cran.r-project.org/package=ggplot2> (65)) and “Umatrix” (<https://cran.r-project.org/package=Umatrix> (31)).

| Sample 1 versus sample 2                  |                     |                                |                        |                    |
|-------------------------------------------|---------------------|--------------------------------|------------------------|--------------------|
| SA1P                                      | Sphingomyelin 42:1  | Palmitic acid 16:0             | Eicosaeinoic acid 20:1 | PE 38:5            |
| LacCeramid C16                            | LPE 22:6            | Margaritic acid 17:0           | 2-AG                   | LPC 22:4           |
| S1P                                       | LPE 18:0p           | Sphingomyelin 42:3             | OEA                    | Cholesterolsulfate |
| Sphingomyelin 36:3                        | LPE 18:0            | LPC 18:0                       | Sphingomyelin 40:1     |                    |
| Ceramide 18:0                             | LPC 20:1            | LPC 18:1                       | Sphingomyelin 42:2     |                    |
| Ceramide 24:0                             | Nervoneic acid 24:1 | Dehydroepiandrosterone sulfate |                        |                    |
| Sample 2: neuropathy versus no neuropathy |                     |                                |                        |                    |
| SA1P                                      | Sphingomyelin 33:1  | Sphingomyelin 43:1             |                        |                    |

**Table 1**

**Lists of lipid mediators that were most informative in assigning a sample (i) to the first or second sampling time point or (ii) a sample from the second time point to a patient with or without neuropathy**

Abbreviations: SA1P: sphinganine-1-phosphate, S1P: sphingosine-1-phosphate, LPE: lysophosphatidylethanolamine, LPC: lysophosphatidylcholine, 2-AG: 2-arachidonoylglycerol, OEA. Oleoylethanolamide.

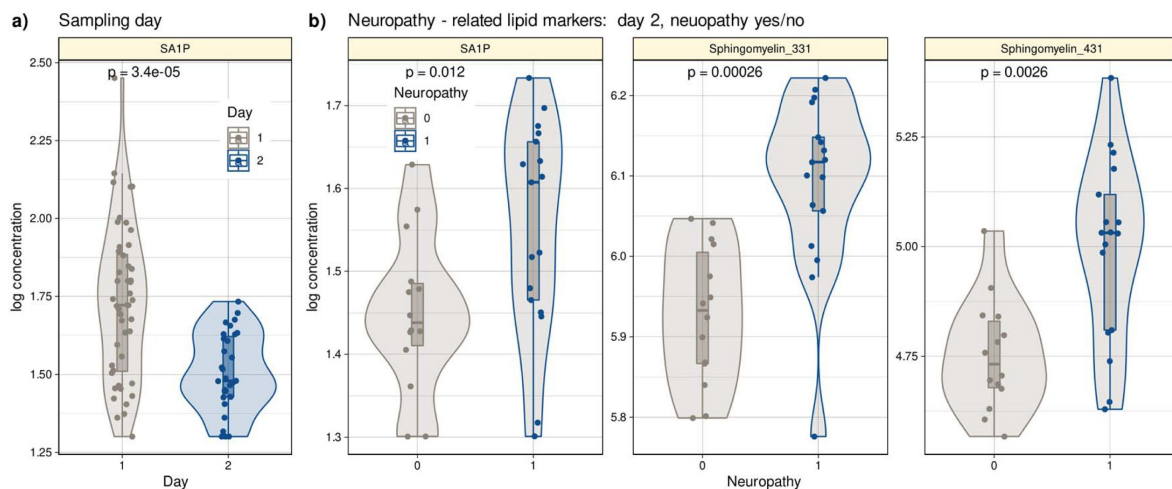


| Classifier                     | Performance                          | Feature set           |                       |                       |                       |
|--------------------------------|--------------------------------------|-----------------------|-----------------------|-----------------------|-----------------------|
|                                | measure                              | Full                  | Reduced               | Reduced<br>permuted   | Sparse                |
|                                | <b>Number of lipid<br/>mediators</b> | 255                   | 77                    | 77                    | 27                    |
| <b>SVM</b>                     | Balanced                             | 0.7 (0.48 -           | 0.78 (0.61 -          | 0.48 (0.23 -          | 0.75 (0.54 -          |
|                                | accuracy                             | 0.92)                 | 1)                    | 0.76)                 | 0.91)                 |
| <b>Random<br/>forests</b>      |                                      | 0.7 (0.56 -<br>0.83)  | 0.75 (0.58 -<br>0.85) | 0.46 (0.24 -<br>0.75) | 0.74 (0.55 -<br>0.88) |
| <b>Logistic<br/>regression</b> |                                      | 0.7 (0.52 -<br>0.85)  | 0.77 (0.58 -<br>0.92) | 0.48 (0.29 -<br>0.7)  | 0.7 (0.49 -<br>0.89)  |
| <b>SVM</b>                     | roc-auc                              | 0.88 (0.67 -<br>1)    | 0.95 (0.85 -<br>1)    | 0.48 (0.16 -<br>0.77) | 0.9 (0.81 -<br>0.99)  |
| <b>Random<br/>forests</b>      |                                      | 0.86 (0.76 -<br>0.95) | 0.88 (0.81 -<br>0.98) | 0.48 (0.21 -<br>0.81) | 0.9 (0.8 - 1)         |
| <b>Logistic<br/>regression</b> |                                      | 0.81 (0.64 -<br>1)    | 0.88 (0.75 -<br>1)    | 0.46 (0.16 -<br>0.79) | 0.86 (0.69 -<br>0.98) |

**Table 2**

### Internal validation of the sets of lipid mediators resulting from the feature selection analysis

The different classifiers (linear support vector machine, SVM, random forests, and logistic regression) were trained with subsets of the training data set with all variables ( $d = 255$  lipid mediators as “full” feature set and with the  $d = 77$  or  $d = 27$  lipid mediators that had resulted from the recursive cABC analysis applied on the sum score of selections by 17 different feature selection methods as “reduced” or “sparse” feature sets, respectively. The trained classifiers were applied to a validation sample comprising 20% of the data that had been removed in a class-proportional manner from the dataset at the beginning of feature selection and had not been touched until used in the classifier validation task presented in this table. In addition, the validation task was repeated with training the classifiers with permuted lipid mediators to observe possible overfitting. Shown are the medians and nonparametric 95% confidence intervals (2.5<sup>th</sup> to 97.5<sup>th</sup> percentiles) from 5 x 20 nested cross-validation runs. Results of external validation in an independent cohort are shown in the supporting information (Table S3).



**Figure 3**

### Log<sub>10</sub>-transformed concentrations of lipid mediators shown to be informative for assigning a post-therapy sample to a patient with pathy or a patient without neuropathy

Individual data points are presented as dots on violin plots showing the probability density distribution of thees, overlaid with box plots where the boxes were constructed using the minimum, quartiles, median (solid line inside the box) and maximum of these. The whiskers add 1.5 times the interquartile range (IQR) to the 75<sup>th</sup> percentile or subtract 1.5 times the IQR from the 25<sup>th</sup> percentile. **a)**: Concentrations of (top hit for sample 1 versus sample 2 segregation) are presented separately for the first and second samples. **b)**: Concentrations of the top lipid mediators uropathy versus no neuropathy in the second sample presented separately for neuropathy-positive and -negative samples. The results of the group rison statistics are given at the top of the graphs. The figure has been created using the R software package (version 4.1.2 for Linux; <http://CRAN.R-project.org/> (26)) and the R library "ggplot2" (<https://cran.r-project.org/package=ggplot2> (65)).

## Biological in-vitro validation of the machine learning based results

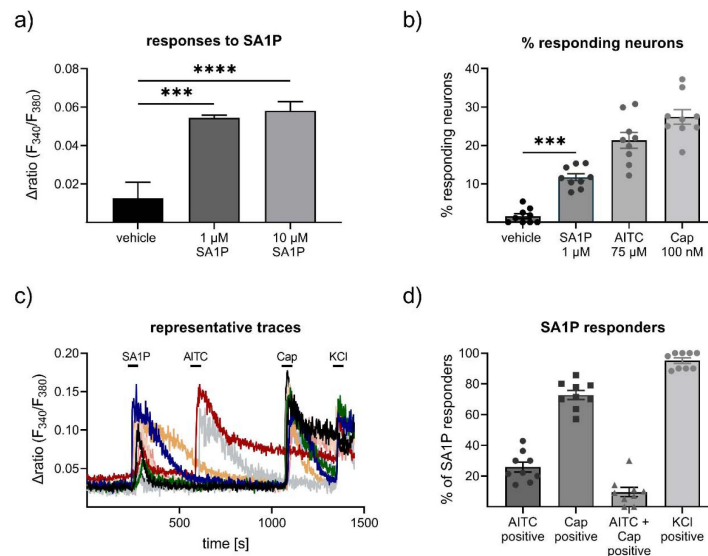
The results of the supervised analysis thus established a limited set of lipid mediators to be regulated in association with paclitaxel therapy or with its side effect of inducing neuropathy. The top hit was sphinganine-1-phosphate (SA1P), also known as dihydrosphingosine-1-phosphate (DH-S1P), providing a basis for in vitro validation of its biological effects in the present context of neuropathy.

Calcium imaging measurements were performed on primary sensory neurons obtained from the murine dorsal root ganglia. We stimulated the neurons with 1 and 10  $\mu\text{M}$  SA1P (ranked as the primary hit) or LPC 24:0 (ranked as one of the least relevant lipids by machine learning analysis). We observed that SA1P caused a direct calcium transient in approximately 11.7% of KCl-responsive sensory neurons (**Figure 4a**, b). However, LPC 24:0 did not induce any notable activation of sensory neurons at concentration of 1 and 10  $\mu\text{M}$  (Figure S4a-c). To further characterize the SA1P-responding neurons, we investigated their responsiveness to agonists of the TRP channels TRPV1 (capsaicin) and TRPA1 (AITC, allyl isothiocyanate), both of which are hallmarks of subpopulations of primary sensory neurons (46). Stimulating SA1P-responsive neurons with capsaicin and AITC revealed that 73% of these neurons also responded to capsaicin and 25% of them responded to AITC, whereas only 9.6% responded to both stimuli. Neurons were identified as responders to KCl (50 mM, 1 min; **Figure 4c, d**).

To identify the receptors or channels responsible for SA1P-mediated calcium transients in sensory neurons, the selective TRPV1 antagonist AMG9810 was used. Neurons were stimulated twice with SA1P (1  $\mu\text{M}$ , 1 min) and AMG9810 (1  $\mu\text{M}$  or vehicle) was added two minutes prior to the second SA1P stimulus. The second SA1P response was entirely abolished when the neurons were treated with AMG9810, but not with the vehicle (**Figure 5a-c**). The potency of AMG9810 was validated using the same measurement protocol as before but with capsaicin (200 nM, 20s) instead of SA1P, which is the gold standard agonist of TRPV1. AMG9810 completely blocked the second capsaicin response (**Figure 5d**). The involvement of S1P-receptors previously suggested to be the receptors for SA1P (47), was evident by studying S1PR1 and S1PR3 as the most highly expressed S1P receptors in sensory neurons (48), which are also targets of the approved drug fingolimod. Sensory neurons were incubated with fingolimod or vehicle for one hour and stimulated the neurons with SA1P (**Fig. 5e, f**). Comparing fingolimod- and vehicle-treated neurons, we observed that the response intensity to SA1P was similar (**Figure 5g**), while the number of neurons responding to SA1P was significantly decreased after fingolimod treatment (**Figure 5h**).

## Discussion

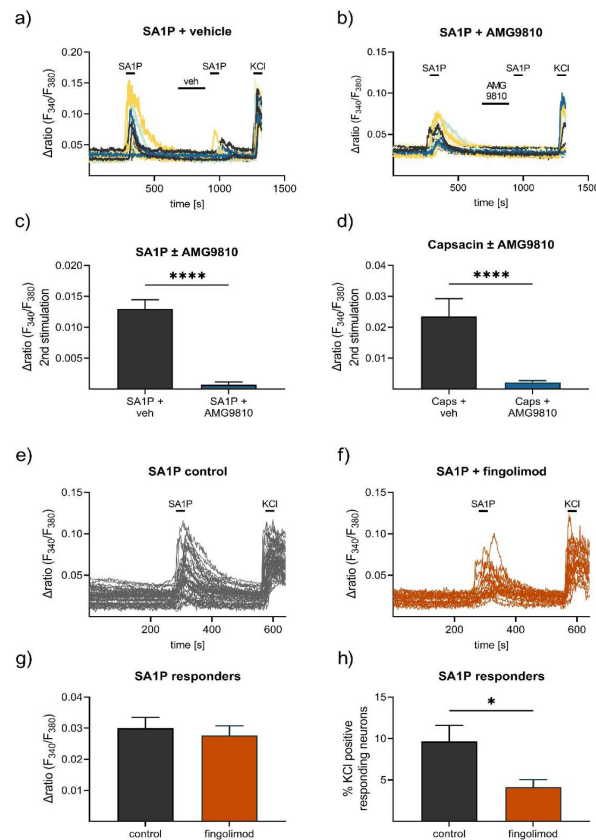
SA1P induced a direct calcium transient in sensory neurons dependent on sphingosine 1-phosphate receptors (S1PR) and the transient receptor potential vanilloid 1 (TRPV1) channel. The results suggest that lipids are altered during paclitaxel treatment and that alterations in sphingolipid metabolism may be critical for the development of paclitaxel-induced peripheral neuropathy in patients. The final (“sparse”) set of lipids regulated between sampling days (before and after paclitaxel treatment) was enriched for sphingolipids. Specifically, sphingolipids were significantly overrepresented among the top hits of lipids regulated between sampling days (Fisher’s exact test:  $p = 0.01$ ), i.e., while 46 of the original 255 lipid mediators were sphingolipids (18%) (Table S2), 11 of the 27 members of the final sparse marker set (40.7%) belonged to the group of sphingolipids. The main pathway in which the top hits are involved is shown in **Figure 6**. With the top hits (“sparse” feature set), three different algorithms (logistic regression, random forests, support vector machines) could be trained to identify, in new cases, whether a blood sample was drawn before or after paclitaxel therapy with a median balanced accuracy of up to



**Figure 4**

### Effects of sphinganine-1-phosphate on primary sensory neurons

a) Neurons were stimulated with SA1P (1 or 10  $\mu\text{M}$ , 1min or vehicle (0.7% methanol (v/v)). b) percentage of responding neurons to vehicle (0.7% methanol (v/v), 1 min), (SA1P (1  $\mu\text{M}$ , 1min), AITC (allyl isothioncyanate, 75  $\mu\text{M}$ , 30s) or capsaicin (caps, 200 nM, 20s). c) representative traces of SA1P-responding neurons and their response to AITC, capsaicin and KCl. d) percentage of SA1P-responding neurons responding to AITC, capsaicin (caps), AITC and capsaicin and KCl. Data are shown as mean  $\pm$  SEM from at least six measurements per condition with at least 40 neurons per measurement, \*  $p < 0.05$ , \*\*  $p < 0.01$ , \*\*\*  $p < 0.001$ , One-way ANOVA



**Figure 5**

### Contribution of TRPV1 and S1P receptors to SA1P-mediated calcium-influx in sensory neurons

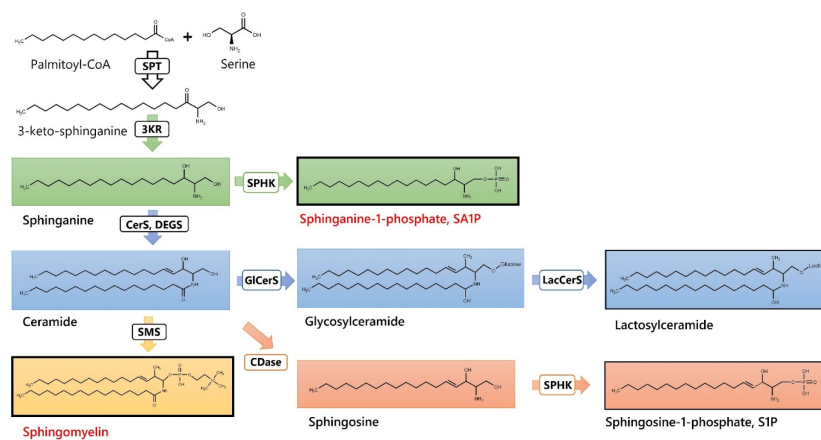
Sensory neurons were stimulated with SA1P twice (1  $\mu$ M, 1 min) and either **a)** vehicle (DMSO 0.003% (v/v), 2 min) or **b)** the TRPV1 antagonist AMG9810 (1  $\mu$ M, 2 min) prior to the second SA1P stimulus. Cells were depolarized with KCl (50 mM, 1 min) at the end of each experiment. **c)** Statistical analysis of the amplitude of SA1P-mediated calcium transients in sensory neurons treated with either vehicle or AMG9810 (blue). **d)** Statistical analysis of the amplitude of capsaicin-mediated calcium transients (100 nM, 20s) in sensory neurons treated with either vehicle or AMG9810 (blue). **e), f)** Sensory neurons were stimulated with SA1P after preincubation with the S1P1 receptor modulator fingolimod (1  $\mu$ M, 1 h) or control. **g)** Statistical analysis of the amplitude of SA1P-mediated calcium transients (1  $\mu$ M, 1min) in sensory neurons treated with either vehicle or fingolimod (1  $\mu$ M, 1h, orange). **h)** Statistical analysis of the number of SA1P-responding neurons (as % of KCl-positives) after treatment with either vehicle or fingolimod (1  $\mu$ M, 1h, orange). Data represents mean  $\pm$  SEM from at least five measurements per condition with at least 25 neurons per measurement, \*  $p < 0.05$ , \*\*  $p < 0.01$ , \*\*\*  $p < 0.01$ , Student's t-test with Welch's correction.

90%. Furthermore, the classification of patients with regard to neuropathy after paclitaxel treatment was reflected in lipidomics in another sphingolipid, i.e., dihydro sphingosine sphinganine-1-phosphate (SA1P), which was elevated in patients with neuropathy.

The results from an unbiased machine-learning-based analysis are in line with those of previous reports that had used classical statistics mainly. For example, S1P was elevated in the spinal cord of mice after bortezomib treatment and during bortezomib-induced neuropathic pain. Blocking S1P1 receptor S1P1R with fingolimod effectively reduced bortezomib-induced mechanical hypersensitivity in vivo (14). Interestingly, targeting the S1P-S1P1R-axis was also found to reduce paclitaxel-induced neuropathic pain in vivo in a preclinical study (49). Further agreements of the present results relate to previous preclinical reports highlighting the significance of the sphingolipid pathway in persistent and neuropathic pain states (14, 50, 51). In addition, the S1P signaling axis was observed to be relevant in neuropathy and chemotherapy-induced neuropathic pain, which led to the suggestion of targeting S1P receptors as a novel approach to ameliorate chemotherapy-induced neuropathy and neuropathic pain (49, 52–55). Taken together, present results from lipid screening and unbiased machine-learning approach point towards a significant contribution of sphingolipid signaling in paclitaxel-induced neuropathy in patients, mainly via sphinganine-1-phosphate and sphingomyelins 33:1 and 43:1.

Sphinganine is a key branching point in the sphingolipid pathway, where it can either be acylated to form dihydroceramides or phosphorylated to SA1P by sphingosine kinases (56). Accumulation of sphinganine has been previously associated with reduced activity of ceramide synthase CerS2 (57). Interestingly, low CerS2 expression is a hallmark of various tumors (58). It is conceivable that a subgroup of the patient cohort still exhibits low CerS2 expression after paclitaxel treatment, which is associated with higher SA1P levels and a higher occurrence of neuropathy. We also identified SA1P as a potential proalgesic lipid mediator, as it causes direct calcium transients in approximately 10% of sensory neurons. This effect is mediated, at least in part, by S1P receptors S1P1 and S1P3 and the TRPV1 channel. The TRPV1 channel has previously been identified as an important mediator of neuropathic pain and is associated with exacerbated activity of sensory neurons during paclitaxel-induced neuropathic pain (59–61). Our data suggest that an S1P receptor modulator, such as fingolimod may be a potential treatment strategy for reducing the proalgesic effect of SA1P and possibly for reducing paclitaxel-induced neuropathy in patients.

Several other lipids were found in the extended list of hits previously associated with chemotherapy-induced neuropathy or acute pain in preclinical studies, such as LPC 18:1, sphingosin-1-phosphate (S1P), and 9,10-EpOME. LPC 18:1 has previously been identified as an endogenous activator of TRPV1 and TRM8 and was found at elevated levels in murine DRGs 24 h after oxaliplatin treatment. This may contribute to oxaliplatin-induced acute pain (62). Similarly, 9,10-EpOME was elevated in the DRGs of paclitaxel-treated mice. Inhibiting its synthesis with the CYP2J2-inhibitor and the approved drug telmisartan was shown to reduce acute paclitaxel-induced mechanical hypersensitivity in vivo and prevent paclitaxel-induced mechanical allodynia by pretreatment (13). In addition, the direct TRPV1 agonist LPA 18:1 was found in the extended list of hits. Lipid was shown to bind to the C-terminal binding site of TRPV1 to increase the opening probability of the channel (63). Other signaling lipids with potential proalgesic effects and potential TRP channel activators that have been described previously have also been identified as crucial for group separation by our unbiased machine-learning approach, which strengthens the presumption that the identified lipids may indeed be connected with paclitaxel neurotoxicity and paclitaxel-induced neuropathy in patients. Additionally, we found several precursor fatty acids in the extended list of hits, including arachidonic acid, linoleic acid, and palmitic acid. These results imply a major dysregulation of lipids after paclitaxel treatment, leading to enhanced plasma levels of precursors for eicosanoids and oxidized linoleic acid metabolites, which may explain the observed enhanced concentrations of the eicosanoids of 5-HETE, 5,6-DHET, and the oxidized linoleic acid metabolites 9,10-EpOME, 9- and 13-HODE.



**Figure 6**

### Sphingolipids and Ceramides

(SPT: Serine palmitoyl-transferase; 3KR: 3-ketosphinganine reductase; SPHK: Sphingosine kinase; CerS: Ceramide synthase; DEGS: Dihydroceramide desaturase; GICerS: Glucosylceramide synthase; LacCerS: Lactosylceramide synthase; SMS: Sphingomyelin synthase; CDase: Ceramidase). Structures were drawn with ChemDraw 20.



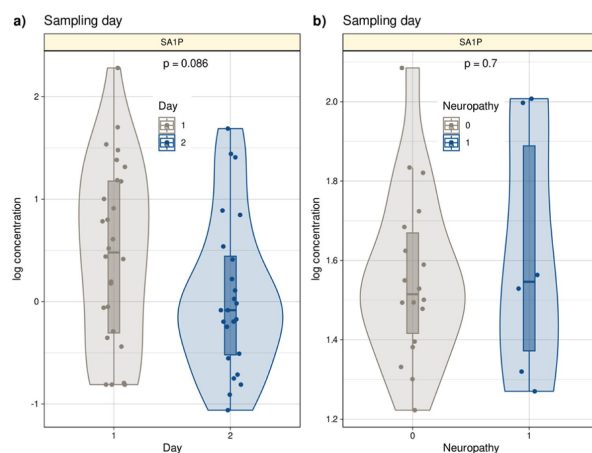
Several limitations need to be addressed. First, the cohort size ( $n=60$  patients,  $n=31$  patients who provided blood samples before and after paclitaxel chemotherapy) of our patient cohort allowed for errors due to interindividual differences. Second, the paclitaxel treatment regimen differed in some patients included in this study. While the majority of the patients received paclitaxel as “pacli weekly” regime consisting of a weekly dose of  $80 \text{ mg/m}^2$  for 12 consecutive weeks in, some few patients received paclitaxel doses up to  $220 \text{ mg/m}^2$  or a combination chemotherapy consisting of carboplatin/paclitaxel. These variabilities within the patient cohort hamper interindividual correlations between plasma lipid mediator concentrations and paclitaxel-induced neuropathy throughout the treatment course. Third, the assessment of peripheral neuropathy in patients was performed according to the guidelines of the NCI Common Terminology Criteria for Adverse Events (CTCAE) v5.0, which ranks the severity of neuropathy into five grades but is rather focused on general adverse events of chemotherapy rather than specifically assessing peripheral neuropathy in detail. We did not perform any neurological testing of sensory parameters, such as quantitative sensory testing (QST) (64 [↗](#)), to determine the sensory status quo of the patients. The study lacked a separate validation cohort with similar data to the main cohort. However, an independent second cohort was available from another hospital (Oncological Center in Offenbach, Germany). This cohort consisted of 28 patients treated with the “paclitaxel weekly” regimen (paclitaxel  $80 \text{ mg/m}^2$ , once weekly) as adjuvant or neoadjuvant therapy for breast or ovarian cancer. All patients provided informed consent into study participation and publication of the results. Blood samples were available from routine collections and were analyzed by LC-MS/MS (Figure S5, Table S3). In contrast to the main cohort, plasma from patients in the second cohort was collected after 6 cycles of paclitaxel treatment due to local routines. Therefore, the second cohort cannot be considered a state-of-the-art validation cohort. In addition, only six patients in this cohort had neuropathy after chemotherapy (26.5%), all with grade 1 neuropathy (Table S4). Despite these limitations, algorithms trained with lipid information from cohort 1 were able to successfully identify whether a probe was taken before or after paclitaxel therapy in the second cohort at a better than guessing level. Furthermore, a trend towards different SA1P concentrations in the plasma of patients after paclitaxel treatment was observed ( $p = 0.086$ , **Figure 7** [↗](#)).

## Conclusions

Here, we demonstrate that the combination of state-of-the-art lipidomics using LC-MS/MS, LC-QTOF-MS, and machine learning-based data analysis can robustly lead to the generation of testable hypotheses and the identification of biologically relevant signaling mediators of neuropathy in an unbiased manner. Lipidomic profiles were compared within the same patients, allowing analysis of individual paclitaxel-induced lipidome changes in the same patients. These analyses led to the identification of a lipid mediator that can directly activate calcium transients in sensory neurons, thereby modulating nociceptive processing and sensory neuron activity. The identified SA1P, through its receptors, provides a potential drug target for co-therapy with paclitaxel to reduce one of its major and therapy-limiting side effects.

## Data availability

Data are available on request from the senior author. Relevant Python and R code is available at <https://github.com/JornLotsch/PaclitaxelNeuropathyProject> [↗](#).



**Figure 7**

### Log10-transformed concentrations of SA1P in the second patient cohort

Individual data points are presented as dots on violin plots showing the probability density distribution of the variables, overlaid with box plots where the boxes were constructed using the minimum, quartiles, median (solid line inside the box) and maximum of these values. The whiskers add 1.5 times the interquartile range (IQR) to the 75th percentile or subtract 1.5 times the IQR from the 25th percentile. a): Concentrations of SA1P (top hit for sample 1 versus sample 2 segregation) are presented separately for the first and second samples. b): Concentrations of S1AP in the second sample are shown separately for neuropathy-positive and -negative samples. Day 1 represents the timepoint before starting chemotherapy. Day 2 represents the timepoint after 12 cycles of paclitaxel chemotherapy. The results of the t-test group comparison statistics are given at the top of the graphs. The figure has been created using the R software package (version 4.1.2 for Linux; <http://CRAN.R-project.org/> (26)) and the R library "ggplot2" (<https://cran.r-project.org/package=ggplot2> (65)).

## Author Contributions

MS and GG conceptualized and designed the experiments. JL and SM performed the machine learning analysis. LH, CA, YS, ST, DT, and NFB performed lipidomics measurements and analyzed the data. SW and MS performed calcium imaging experiments. KG, CB, and CS assembled the patient cohorts. JL, KG, SW and MS wrote the manuscript. All authors have reviewed and edited the manuscript.

## Acknowledgements

MS was supported by the Deutsche Forschungsgemeinschaft (German Research Foundation, DFG, Grants SFB1039 A09 and Z01) and from the Fraunhofer Foundation Project: Neuropathic Pain as well as the Fraunhofer Cluster of Excellence for Immune-Mediated Diseases (CIMD). This work was also supported by the Leistungszentrum Innovative Therapeutics (TheraNova) funded by the Fraunhofer Society and the Hessian Ministry of Science and Arts. JL was supported by the Deutsche Forschungsgemeinschaft (DFG LO 612/16-1). These public funders had no role in study design, data collection and analysis, decision to publish, or preparation of the manuscript. We would also like to thank Drs Tabea Osthues, Béla Zimmer and Vittoria Rimola, as well as Mr. Maksim Sendetski for their help in patient sample processing.

## Supplementary data captions

Figure S1: Log10-transformed concentrations of the analyzed lipid mediators presented as violin plots showing the probability density distribution of the variables; Figure S2: Results of a projection of the z-standardized log-transformed lipidomics data onto a lower-dimensional space by means of PCA; Figure S3: Identification of the lipid mediators that were most informative in assigning a sample to the pre- or post-therapy time point. Feature selection by 13 different methods; Figure S4: Effects of Lysophosphatidylcholine 24:0 (LPC 24:0) on primary sensory neurons; Figure S5: Plasma lipids from the independent second patient cohort. Table S1: Patient characteristics of the 31 patients that gave blood samples before and after chemotherapy from the patient cohort; Table S2: Complete list of lipid mediators included in the analyses, separated by group of lipid and detection method; Table S3: External validation of the classifiers; Table S4: Patient characteristics of the 28 patients from the second cohort that gave blood samples before and after the sixth cycle of chemotherapy. Supplementary methods: data analysis, data preprocessing, supervised and unsupervised machine learning, lipid mediatorinformative for assigning samples to before or after therapy in an independent second patient cohort, lipidomics analyses of plasma patient samples (Tables 1-10).

## References

1. Pachman DR, Barton DL, Watson JC, Loprinzi CL (2011) **Chemotherapy-induced peripheral neuropathy: prevention and treatment** *Clin Pharmacol Ther* **90**:377–87
2. Park SB, Goldstein D, Krishnan AV, Lin CS, Friedlander ML, Cassidy J, et al. (2013) **Chemotherapy-induced peripheral neurotoxicity: a critical analysis** *CA: a cancer journal for clinicians* **63**:419–37
3. Cavaletti G, Marmiroli P (2020) **Management of Oxaliplatin-Induced Peripheral Sensory Neuropathy** *Cancers (Basel)* **12**
4. Smith EML, Pang H, Cirrincione C, Fleishman S, Paskett ED, Ahles T, et al. (2013) **Effect of Duloxetine on Pain, Function, and Quality of Life Among Patients With Chemotherapy-Induced Painful Peripheral Neuropathy A Randomized Clinical Trial** . *Jama-J Am Med Assoc* **309**:1359–67
5. Gornstein E, Schwarz TL (2014) **The paradox of paclitaxel neurotoxicity: Mechanisms and unanswered questions** *Neuropharmacology* **76**:175–83
6. Hershman DL, Lacchetti C, Dworkin RH, Lavoie Smith EM, Bleeker J, Cavaletti G, et al. (2014) **Prevention and management of chemotherapy-induced peripheral neuropathy in survivors of adult cancers: American Society of Clinical Oncology clinical practice guideline** *J Clin Oncol* **32**:1941–67
7. Yang CH, Horwitz SB (2017) **Taxol((R)): The First Microtubule Stabilizing Agent** *Int J Mol Sci* **18**
8. Diaz PL, Furfari A, Wan BA, Lam H, Charames G, Drost L, et al. (2018) **Predictive biomarkers of chemotherapy-induced peripheral neuropathy: a review** *Biomark Med* **12**:907–16
9. Sisignano M, Lotsch J, Parnham MJ, Geisslinger G (2019) **Potential biomarkers for persistent and neuropathic pain therapy** *Pharmacol Ther* **199**:16–29
10. Huehnchen P, Schinke C, Bangemann N, Dordevic AD, Kern J, Maierhof SK, et al. (2022) **Neurofilament proteins as a potential biomarker in chemotherapy-induced polyneuropathy** *Jci Insight* **7**
11. Hohmann SW, Angioni C, Tunaru S, Lee S, Woolf CJ, Offermanns S, et al. (2017) **The G2A receptor (GPR132) contributes to oxaliplatin-induced mechanical pain hypersensitivity** *Sci Rep* **7**
12. Piomelli D, Sasso O (2014) **Peripheral gating of pain signals by endogenous lipid mediators** *Nat Neurosci* **17**:164–74
13. Sisignano M, Angioni C, Park CK, Meyer Dos Santos S, Jordan H, Kuzikov M, et al. (2016) **Targeting CYP2J to reduce paclitaxel-induced peripheral neuropathic pain** *Proc Natl Acad Sci U S A* **113**:12544–9
14. Stockstill K, Doyle TM, Yan X, Chen Z, Janes K, Little JW, et al. (2018) **Dysregulation of sphingolipid metabolism contributes to bortezomib-induced neuropathic pain** *J Exp Med* **215**:1301–13

15. De Aguilar JL Gonzalez (2019) **Lipid Biomarkers for Amyotrophic Lateral Sclerosis** *Front Neurol* **10**
16. Trostchansky A (2019) **Overview of Lipid Biomarkers in Amyotrophic Lateral Sclerosis (ALS)** *Adv Exp Med Biol* **1161**:233–41
17. Zarrouk A, Debbabi M, Bezine M, Karym EM, Badreddine A, Rouaud O, et al. (2018) **Lipid Biomarkers in Alzheimer's Disease** *Curr Alzheimer Res* **15**:303–12
18. Osthues T, Sisignano M (2019) **Oxidized Lipids in Persistent Pain States** *Front Pharmacol* **10**
19. Shapiro H, Singer P, Ariel A (2016) **Beyond the classic eicosanoids: Peripherally-acting oxygenated metabolites of polyunsaturated fatty acids mediate pain associated with tissue injury and inflammation** *Prostaglandins Leukot Essent Fatty Acids* **111**:45–61
20. Sisignano M, Bennett DL, Geisslinger G, Scholich K (2014) **TRP-channels as key integrators of lipid pathways in nociceptive neurons** *Prog Lipid Res* **53**:93–107
21. Brunkhorst-Kanaan N, Klatt-Schreiner K, Hackel J, Schroter K, Trautmann S, Hahnefeld L, et al. (2019) **Targeted lipidomics reveal derangement of ceramides in major depression and bipolar disorder** *Metabolism* **95**:65–76
22. Leclercq M, Vittrant B, Martin-Magniette ML, Scott Boyer MP, Perin O, Bergeron A, et al. (2019) **Large-Scale Automatic Feature Selection for Biomarker Discovery in High-Dimensional OMICs Data** *Front Genet* **10**
23. Khadirnaikar S, Shukla S, Prasanna SRM (2023) **Machine learning based combination of multi-omics data for subgroup identification in non-small cell lung cancer** *Sci Rep* **13**
24. Hu YH, Palreddy S, Tompkins WJ (1997) **A patient-adaptable ECG beat classifier using a mixture of experts approach** *IEEE Trans Biomed Eng* **44**:891–900
25. Ihaka R, Gentleman R (1996) **R: A Language for Data Analysis and Graphics** *Journal of Computational and Graphical Statistics* **5**:299–314
26. R Development Core Team (2008) **R: A Language and Environment for Statistical Computing**
27. Van Rossum G, Drake FL. (1995) **Python tutorial** *Centrum voor Wiskunde en Informatica Amsterdam*
28. Hotelling H (1933) **Analysis of a complex of statistical variables into principal components** *Journal of Educational Psychology* **24**:498–520
29. Pearson K. LIII (1901) **On lines and planes of closest fit to systems of points in space. The London Edinburgh, and Dublin Philosophical Magazine and Journal of Science** **2**:559–72
30. Kohonen T (1982) **Self-organized formation of topologically correct feature maps** *Biol Cybernet* **43**:59–69
31. Lötsch J, Lerch F, Djaldetti R, Tegeder I, Ultsch A (2018) **Identification of disease-distinct complex biomarker patterns by means of unsupervised machine-learning using an interactive R toolbox (Umatrix)** *BMC Big Data Analytics* **3** <https://doi.org/10.1186/s41044-018-0032-1>

32. Ultsch A. (2003) **Maps for Visualization of High-Dimensional Data Spaces** *WSOM* :225–30
33. Ultsch A, Lötsch J (2017) **Machine-learned cluster identification in high-dimensional data** *J Biomed Inform* **66**:95–104
34. Ultsch A, Sieman HP. (1990) **INNC'90, Int Neural Network Conference** :305–8
35. Lötsch J, Ultsch A., Villmann T, Schleif F-M, Kaden M, Lange M (2014) **Advances in Intelligent Systems and Computing** :248–57
36. Cramer JS. (2002) **The Origins of Logistic Regression**
37. Ho TK (1995) **Proceedings of the Third International Conference on Document Analysis and Recognition (Volume 1) - Volume 1** *IEEE Computer Society* **278**
38. Breiman L (2001) **Random Forests** *Mach Learn* **45**:5–32
39. Cortes C, Vapnik V (1995) **Support-Vector Networks** *Machine Learning* **20**:273–97
40. Ultsch A, Lötsch J (2015) **Computed ABC Analysis for Rational Selection of Most Informative Variables in Multivariate Data** *PLoS One* **10**
41. Juran JM (1975) **The non-Pareto principle; Mea culpa** *Quality Progress* **8**:8–9
42. Sisignano M, Park CK, Angioni C, Zhang DD, von Hehn C, Cobos EJ, et al. (2012) **5,6-EET Is Released upon Neuronal Activity and Induces Mechanical Pain Hypersensitivity via TRPA1 on Central Afferent Terminals** *The Journal of neuroscience : the official journal of the Society for Neuroscience* **32**:6364–72
43. Pachman DR, Barton DL, Watson JC, Loprinzi CL (2011) **Chemotherapy-induced peripheral neuropathy: prevention and treatment** *Clin Pharmacol Ther* **90**:377–87
44. Scripture CD, Figg WD, Sparreboom A (2006) **Peripheral neuropathy induced by paclitaxel: recent insights and future perspectives** *Curr Neuropharmacol* **4**:165–72
45. Lötsch J, Ultsch A (2023) **Recursive computed ABC (cABC) analysis as a precise method for reducing machine learning based feature sets to their minimum informative size** *Sci Rep* **13**
46. Julius D (2013) **TRP Channels and Pain** *Annu Rev Cell Dev Biol* **29**:355–84
47. Magaye RR, Savira F, Hua Y, Kelly DJ, Reid C, Flynn B, et al. (2019) **The role of dihydrosphingolipids in disease** *Cell Mol Life Sci* **76**:1107–34
48. Quarta S, Camprubi-Robles M, Schweigreiter R, Matusica D, Haberberger RV, Proia RL, et al. (2017) **Sphingosine-1-Phosphate and the S1P3 Receptor Initiate Neuronal Retraction via RhoA/ROCK Associated with CRMP2 Phosphorylation** *Front Mol Neurosci* **10**
49. Janes K, Little JW, Li C, Bryant L, Chen C, Chen Z, et al. (2014) **The Development and Maintenance of Paclitaxel-induced Neuropathic Pain Require Activation of the Sphingosine 1-Phosphate Receptor Subtype 1** *J Biol Chem* **289**:21082–97
50. Squillace S, Spiegel S, Salvemini D (2020) **Targeting the Sphingosine-1-Phosphate Axis for Developing Non-narcotic Pain Therapeutics** *Trends Pharmacol Sci* **41**:851–67

51. Chen Z, Doyle TM, Luongo L, Largent-Milnes TM, Giancotti LA, Kolar G, et al. (2019) **Sphingosine-1-phosphate receptor 1 activation in astrocytes contributes to neuropathic pain** *Proc Natl Acad Sci U S A*
52. Wang W, Xiang P, Chew WS, Torta F, Bandla A, Lopez V, et al. (2020) **Activation of sphingosine 1-phosphate receptor 2 attenuates chemotherapy-induced neuropathy** *J Biol Chem* **295**:1143–52
53. Salvemini D, Doyle T, Kress M, Nicol G (2013) **Therapeutic targeting of the ceramide-to-sphingosine 1-phosphate pathway in pain** *Trends Pharmacol Sci* **34**:110–8
54. Becker KA, Uerschels AK, Goins L, Doolen S, McQuerry KJ, Bielawski J, et al. (2020) **Role of 1-Deoxysphingolipids in docetaxel neurotoxicity** *J Neurochem* **154**:662–72
55. Chua KC, Xiong C, Ho C, Mushiroda T, Jiang C, Mulkey F, et al. (2020) **Genomewide Meta-Analysis Validates a Role for S1PR1 in Microtubule Targeting Agent-Induced Sensory Peripheral Neuropathy** *Clin Pharmacol Ther* **108**:625–34
56. Merrill AH Jr (2011) **Sphingolipid and glycosphingolipid metabolic pathways in the era of sphingolipidomics** *Chem Rev* **111**:6387–422
57. Pewzner-Jung Y, Brenner O, Braun S, Laviad EL, Ben-Dor S, Feldmesser E, et al. (2010) **A critical role for ceramide synthase 2 in liver homeostasis: II. insights into molecular changes leading to hepatopathy** *J Biol Chem* **285**:10911–23
58. Zhang Q, Wang JY, Yan W, Wang DD, Yang SJ, Zhou SY, et al. (2019) **Clinical and pathological significance of Homo sapiens ceramide synthase 2 (CerS-2) in diverse human cancers** *Biosci Rep* **39**
59. Jardin I, Lopez JJ, Diez R, Sanchez-Collado J, Cantonero C, Albarran L, et al. (2017) **TRPs in Pain Sensation** *Front Physiol* **8**
60. Hara T, Chiba T, Abe K, Makabe A, Ikeno S, Kawakami K, et al. (2013) **Effect of paclitaxel on transient receptor potential vanilloid 1 in rat dorsal root ganglion** *Pain* **154**:882–9
61. Kamata Y, Kambe T, Chiba T, Yamamoto K, Kawakami K, Abe K, et al. (2020) **Paclitaxel Induces Upregulation of Transient Receptor Potential Vanilloid 1 Expression in the Rat Spinal Cord** *Int J Mol Sci* **21**
62. Rimola V, Hahnefeld L, Zhao J, Jiang C, Angioni C, Schreiber Y, et al. (2020) **Lysophospholipids Contribute to Oxaliplatin-Induced Acute Peripheral Pain** *J Neurosci* **40**:9519–32
63. Nieto-Posadas A, Picazo-Juarez G, Llorente I, Jara-Oseguera A, Morales-Lazaro S, Escalante-Alcalde D, et al. (2011) **Lysophosphatidic acid directly activates TRPV1 through a C-terminal binding site** *Nat Chem Biol*
64. Maier C, Baron R, Tolle TR, Binder A, Birbaumer N, Birklein F, et al. (2010) **Quantitative sensory testing in the German Research Network on Neuropathic Pain (DFNS): somatosensory abnormalities in 1236 patients with different neuropathic pain syndromes** *Pain* **150**:439–50
65. Wickham H. (2009) **ggplot2: Elegant Graphics for Data Analysis**



## Article and author information

### Jörn Lötsch

Institute of Clinical Pharmacology, Goethe - University, Theodor Stern Kai 7, 60590 Frankfurt am Main, Germany, Fraunhofer Institute for Translational Medicine and Pharmacology ITMP, and Fraunhofer Cluster of Excellence for Immune Mediated Diseases CIMD, Theodor-Stern-Kai 7, 60596 Frankfurt am Main, Germany

### Khayal Gasimli

Goethe University, Department of Gynecology and Obstetrics, Theodor-Stern-Kai 7, Frankfurt/Main, Germany

### Sebastian Malkusch

Institute of Clinical Pharmacology, Goethe - University, Theodor Stern Kai 7, 60590 Frankfurt am Main, Germany, Fraunhofer Institute for Translational Medicine and Pharmacology ITMP, and Fraunhofer Cluster of Excellence for Immune Mediated Diseases CIMD, Theodor-Stern-Kai 7, 60596 Frankfurt am Main, Germany

### Lisa Hahnefeld

Institute of Clinical Pharmacology, Goethe - University, Theodor Stern Kai 7, 60590 Frankfurt am Main, Germany, Fraunhofer Institute for Translational Medicine and Pharmacology ITMP, and Fraunhofer Cluster of Excellence for Immune Mediated Diseases CIMD, Theodor-Stern-Kai 7, 60596 Frankfurt am Main, Germany

### Carlo Angioni

Institute of Clinical Pharmacology, Goethe - University, Theodor Stern Kai 7, 60590 Frankfurt am Main, Germany

### Yannick Schreiber

Fraunhofer Institute for Translational Medicine and Pharmacology ITMP, and Fraunhofer Cluster of Excellence for Immune Mediated Diseases CIMD, Theodor-Stern-Kai 7, 60596 Frankfurt am Main, Germany

### Sandra Trautmann

Institute of Clinical Pharmacology, Goethe - University, Theodor Stern Kai 7, 60590 Frankfurt am Main, Germany, Fraunhofer Institute for Translational Medicine and Pharmacology ITMP, and Fraunhofer Cluster of Excellence for Immune Mediated Diseases CIMD, Theodor-Stern-Kai 7, 60596 Frankfurt am Main, Germany

### Saskia Wedel

Institute of Clinical Pharmacology, Goethe - University, Theodor Stern Kai 7, 60590 Frankfurt am Main, Germany

### Dominique Thomas

Institute of Clinical Pharmacology, Goethe - University, Theodor Stern Kai 7, 60590 Frankfurt am Main, Germany, Fraunhofer Institute for Translational Medicine and Pharmacology ITMP, and Fraunhofer Cluster of Excellence for Immune Mediated Diseases CIMD, Theodor-Stern-Kai 7, 60596 Frankfurt am Main, Germany

**Nerea Ferreiros Bouzas**

Institute of Clinical Pharmacology, Goethe - University, Theodor Stern Kai 7, 60590 Frankfurt am Main, Germany, Fraunhofer Institute for Translational Medicine and Pharmacology ITMP, and Fraunhofer Cluster of Excellence for Immune Mediated Diseases CIMD, Theodor-Stern-Kai 7, 60596 Frankfurt am Main, Germany

**Christian Brandts**

German Cancer Consortium (DKTK) and German Cancer Research Center (DKFZ), Heidelberg, 69120, Germany, Goethe University, University Cancer Center Frankfurt (UCT), Goethe University Hospital, Frankfurt am Main 60590, Germany

**Benjamin Schnappauf**

Oncology Center, Sana-Klinikum Offenbach, Starkenburgring 66, 63069 Offenbach

**Christine Solbach**

Goethe University, Department of Gynecology and Obstetrics, Theodor-Stern-Kai 7, Frankfurt/Main, Germany

**Gerd Geisslinger**

Institute of Clinical Pharmacology, Goethe - University, Theodor Stern Kai 7, 60590 Frankfurt am Main, Germany, Fraunhofer Institute for Translational Medicine and Pharmacology ITMP, and Fraunhofer Cluster of Excellence for Immune Mediated Diseases CIMD, Theodor-Stern-Kai 7, 60596 Frankfurt am Main, Germany

**Marco Sisignano**

Institute of Clinical Pharmacology, Goethe - University, Theodor Stern Kai 7, 60590 Frankfurt am Main, Germany, Fraunhofer Institute for Translational Medicine and Pharmacology ITMP, and Fraunhofer Cluster of Excellence for Immune Mediated Diseases CIMD, Theodor-Stern-Kai 7, 60596 Frankfurt am Main, Germany

**For correspondence:** [Marco.Sisignano@med.uni-frankfurt.de](mailto:Marco.Sisignano@med.uni-frankfurt.de)

ORCID iD: [0000-0002-7581-0951](https://orcid.org/0000-0002-7581-0951)

**Copyright**

© 2024, Lötsch et al.

This article is distributed under the terms of the [Creative Commons Attribution License](https://creativecommons.org/licenses/by/4.0/), which permits unrestricted use and redistribution provided that the original author and source are credited.

**Editors**

Reviewing Editor

**Pramod Mistry**

Yale University, New Haven, United States of America

Senior Editor

**Pramod Mistry**

Yale University, New Haven, United States of America

**Reviewer #1 (Public Review):****Summary:**

This study examines lipid profiles in cancer patients treated with the neurotoxic chemotherapy paclitaxel. Multiple methods, including machine learning as well as more conventional statistical modelling, were used to classify lipid patterns before and after paclitaxel treatment and in conjunction with neuropathy status. Lipid profiles before and after paclitaxel therapy were analysed from 31 patients. The study aimed to characterize from the lipid profile if plasma samples were collected pre-paclitaxel or post-paclitaxel and their relevance to neuropathy status. Sphingolipids including sphinganine-1-phosphate (SA1P) differed between patients with and without neuropathy. To examine the potential role of SA1P, it was applied to murine primary sensory neuron cultures, and produced calcium transients in a proportion of neurons. This response was abolished by the application of a TRPV1 antagonist. The number of neurons responding to SA1P was partially reduced by the sphingosine 1-phosphate receptor (S1PR1) modulator fingolimod.

**Strengths:**

The strengths of this study include the use of multiple methods to classify lipid patterns and the attempt to validate findings from the clinical cohort in a preclinical model using primary sensory neurons.

**Weaknesses:**

There are a number of weaknesses in the study. The small sample size is a significant limitation of the study. Out of 31 patients, only 17 patients were reported to develop neuropathy, with significant neuropathy (grade 2/3) in only 5 patients. The authors acknowledge this limitation in the results and discussion sections of the manuscript, but it limits the interpretation of the results. Also acknowledged is the limited method used to assess neuropathy.

Potentially due to this small number of patients with neuropathy, the machine learning algorithms could not distinguish between samples with and without neuropathy. Only selected univariate analyses identified differences in lipid profiles potentially related to neuropathy.

Three sphingolipid mediators including SA1P differed between patients with and without neuropathy at the end of treatment. These sphingolipids were elevated at the end of treatment in the cohort with neuropathy, relative to those without neuropathy. However, across all samples from pre to post-paclitaxel treatment, there was a significant reduction in SA1P levels. It is unclear from the data presented what the underlying mechanism for this result would be. If elevated SA1P is associated with neuropathy development, it would be expected to increase in those who develop neuropathy from pre to post-treatment time points.

Primary sensory neuron cultures were used to examine the effects of SA1P application. SA1P application produced calcium transients in a small proportion of sensory neurons. It is not clear how this experimental model assists in validating the role of SA1P in neuropathy development as there is no assessment of sensory neuron damage or other hallmarks of peripheral neuropathy. These results demonstrate that some sensory neurons respond to SA1P and that this activity is linked to TRPV1 receptors. However, further studies will be required to determine if this is mechanistically related to neuropathy.

**Impact:**

Taken in total, the data presented do not provide sufficient evidence to support the contention that SA1P has an important role in paclitaxel-induced peripheral neuropathy. Further, the results do not provide evidence to support the use of S1PR1 receptor antagonists as a therapeutic strategy. It is important to be careful with language use in the discussion, as the significance of the present results is overstated.

However, based on the results of previous studies, it is likely that sphingolipid metabolism plays a role in chemotherapy-induced peripheral neuropathy. Based on this existing evidence, the S1PR1 receptor antagonist fingolimod has already been examined in experimental models and clinical trials. Further work is needed to examine the links between lipid mediators and neuropathy development and identify additional strategies for intervention.

<https://doi.org/10.7554/eLife.91941.1.sa1>

#### **Reviewer #2 (Public Review):**

##### **Summary:**

The study investigates the mechanisms underlying chemotherapy-induced peripheral neuropathy (CIPN), a notable side effect of commonly used anticancer drugs like paclitaxel. It aims to comprehend the putative mechanisms through lipidomics analysis of plasma samples from cancer patients pre and post-paclitaxel treatment, drawing inspiration from preclinical studies highlighting the role of sphingolipids. While the use of patient plasma samples stands out as a major strength, shortcomings in the result presentation undermine the study's significance. The introduction lacks a robust rationale, failing to articulate the utility of machine learning methods over conventional lipidomics analysis and the relevance of broader neuropathy in the context of the study's goal of investigating peripheral neuropathy. The failure to robustly link neuropathy to paclitaxel treatment, with only around 50% of patients developing neuropathy, mostly at Grade 1, with no or mild symptoms that require no intervention, weakens the study's impact. The presentation of results lacks clarity on sphingolipid dysregulation, leaving uncertainty regarding downregulation or upregulation. Furthermore, no clarity in validation for the machine learning-based analysis with conventional methods and an overall weakness in result representation weaken the study, despite addressing an important question in the field.

##### **Strengths:**

The study leverages patient plasma samples before and after paclitaxel treatment, enhancing the translatability of findings to patient impact. The attempt to employ machine learning (ML) methods for analyzing biological samples and classifying patient groups is commendable, pushing the biomedical sciences towards ML applications for handling complex data. The chosen topic of investigating chemotherapy-induced peripheral neuropathy (CIPN) is clinically important, offering potential benefits for cancer patients undergoing chemotherapy treatment.

##### **Weaknesses:**

The article is poorly written, hindering a clear understanding of core results. While the study's goals are apparent, the interpretation of sphingolipids, particularly SA1P, as key mediators of paclitaxel-induced neuropathy lacks robust evidence. The introduction fails to establish the significance of general neuropathy or peripheral neuropathy in anticancer drug-treated patients, and crucial details, such as the percentage of patients developing general neuropathy or peripheral neuropathy, are omitted. This omission is particularly relevant given that only around 50% of patients developed neuropathy in this study, primarily of mild Grade 1 severity with negligible symptoms, contradicting the study's

assertion of CIPN as a significant side effect. The lack of clarity in distinguishing results obtained by lipidomics using machine learning methods and conventional methods adds to the confusion. The poorly written results section fails to specify SA1P's downregulation or upregulation, and the process of narrowing down to sphingolipids and SA1P is inadequately explained. Integrating a significant portion of the discussion section into the results section could enhance clarity. An explanation of the utility of machine learning in classifying patient groups over conventional methods and the citation of original research articles, rather than relying on review articles, may also add clarity to the usefulness of the study.

<https://doi.org/10.7554/eLife.91941.1.sa0>